TCC70xx MCU BSP

API Specification

for Analog Digital Converter Controller

Rev. 0.10 [A]

2021-09-03

Preliminary version

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1 INTRODUCTION

This document describes the analog digital converter controller device driver (hereinafter referred to as ADC driver) of TCC70xx MCU BSP. For more detailed information on the analog digital converter controller in the TCC70xx MCU Sub-system, refer to Chapter 15 Analog Digital Converter (ADC) Controller in "*TCC70xx Full Specification*" [1].

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

ADC	Analog Digital Converter

3 SOURCE FILES

3.1 Location of Source Files

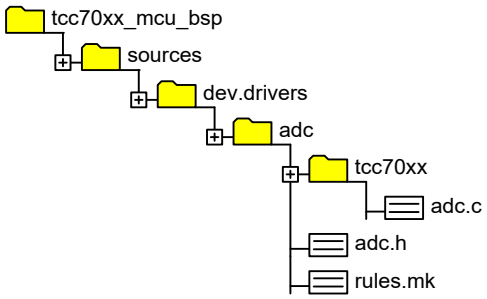

Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- adc.c
  - Source of ADC driver
- adc.h
  - API header of ADC driver
- rules.mk
  - Rules used for build

4 ENUMERATIONS

4.1 ADCChannel_t

Enumeration variables for ADC channel value of ADC user interface API. TCC70xx ADC supports 12 channels.

Declaration

```
typedef enum __ADC_CHANNEL__
{
    ADC_CHANNEL_0      = 0,
    ADC_CHANNEL_1      = 1,
    ADC_CHANNEL_2      = 2,
    ADC_CHANNEL_3      = 3,
    ADC_CHANNEL_4      = 4,
    ADC_CHANNEL_5      = 5,
    ADC_CHANNEL_6      = 6,
    ADC_CHANNEL_7      = 7,
    ADC_CHANNEL_8      = 8,
    ADC_CHANNEL_9      = 9,
    ADC_CHANNEL_10     = 10,
    ADC_CHANNEL_11     = 11,
    ADC_CHANNEL_MAX    = 12
} ADCChannel_t;
```

Description

Value	Description
ADC_CHANNEL_xx	Channel number value for each ADC channel
ADC_CHANNEL_MAX	The maximum number of channels

Remark

None

4.2 ADCMode_t

Enumeration variables for selecting ADC mode

Declaration

```
typedef enum __ADC_MODE___
{
    ADC_MODE_NORMAL    = 0,
    ADC_MODE_IRQ       = 1,
    ADC_MODE_DMA       = 2
} ADCMode_t;
```

Description

Value	Description
ADC_MODE_NORMAL	The conversion value can be read in the ADC sampled data register (SD_OUT0 to SD_OUT11).
ADC_MODE_IRQ	IRQ interrupt is generated when the conversion operation is completed.
ADC_MODE_DMA	The conversion value can be read in the designated location of the DMA.

Remark

None

5 CONSTANTS

5.1 ADC Module

Constant values for TCC70xx ADC Hardware Module.

Declaration

```
#define ADC_MODULE_0  (0)
#define ADC_MODULE_1  (1)
```

Description

Value	Description
ADC_MODULE_0	ADC0 Physical Device Number
ADC_MODULE_1	ADC1 Physical Device Number

5.2 Configurations

Constant values for configuration value for API

Declaration

```
#define ADC_CONV_TIMEOUT  (2UL)
```

Description

Value	Description
ADC_CONV_TIMEOUT	Timeout for Done bit. It waits to set done bit to 1 during ADC_CONV_TIMEOUT * ADC_CONV_CYCLE cycles in **ADC_Read** API.

5.3 Error Numbers

Constant values for error numbers

Declaration

```
#define ADC_ERR_INVALID_MODULE   (-1)
```

Description

Value	Description
ADC_ERR_INVALID_MODULE	Invalid module number is passed into **ADC_Init** API.

6 APIs

6.1 ADC_Init

Initializes the ADC. Regardless of the input type, the ***ADC_Init()*** function performs ADC clock enable.

There are three types of modes in ADC as follows:

- Normal mode: Default mode
- IRQ mode: When conversion is completed, this mode can generate an interrupt.
- DMA mode: By using DMA, this mode can read the conversion result.

Declaration

```
void ADC_Init
(
    uint8   Type,
    uint8   ucModule
);
```

Parameters

Value	*Description*
Type	*[in] ADC Mode (Normal, IRQ, DMA). Refer to ADCMode_t.*
ucModule	*[in] ADC Module number. 0 for ADC0, 1 for ADC1*

Return

None

Remark

None

6.2 ADC_Read

Starts the ADC conversion using the ***ADC_SMP_CMD*** register for single channel, checks the conversion done bit, and reads the conversion result value from the ***SDOUT*** register. This function does not support thread safe and reentrancy on the same channel.

Declaration

```
uint32 ADC_Read
(
    ADCChannel_t   channel,
    uint8          ucModule,
    int32          siWatch
);
```

Parameters

Value	*Description*
channel	*[in] ADC Channel number. Refer to ADCChannel_t.*
ucModule	*[in] ADC Module Number. 0 for ADC0, 1 for ADC1.*
siWatch	*[in] Enable monitoring mode for debugging. It must be 0 by default.*

Return

Value	*Description*
uint32	*ADC conversion value, conversion value range is 0 to 4096 (12 bits).*

Remark

None

6.3 ADC_AutoScan

Calls ***ADC_Read()*** function internally to convert input analog signal for all ADC channels and to read the conversion values one by one. This function does not implement consecutive conversion and not support thread safe and reentrancy. This function must be used in only one task of the all tasks such as user tasks and system tasks.

Declaration

```
uint32 * ADC_AutoScan
(
    uint8   ucModule,
    int32   siWatch
);
```

Parameters

Value	Description
ucModule	[in] ADC Module Number. 0 for ADC0, 1 for ADC1.
siWatch	[in] Enable monitoring mode for debugging. It must be 0 by default.

Return

Value	Description
uint32 *	Static array address storing all channel conversion values

Remark

None

7 How to use ADC Driver

7.1 Initialize ADC Driver

```
void BSP_Init()
{
    …
    ADC_Init(ADC_MODE_NORMAL, ADC_MODULE_0); //ADC Normal Mode for ADC0 initialization
    ADC_Init(ADC_MODE_NORMAL, ADC_MODULE_1); //ADC Normal Mode for ADC1 initialization
    …
}
```

Note: Refer to sources/dev.driver/common/tcc70xx/bsp.c

7.2 Read Converted Value

```
uint32 * ADC_AutoScan
(
    uint8                           ucModule,
    int32                           siWatCh
)
{
    uint32 ch;
    uint32 read_data;
    static uint32 adc_scan_val[ADC_CHANNEL_MAX];

    read_data  = 0;

    for (ch = 0 ; ch < (uint32)ADC_CHANNEL_MAX ; ch++) // read every channels for ADC module
    {
        read_data          = ADC_Read((ADCChannel_t)ch, ucModule, siWatCh);
        adc_scan_val[ch]   = read_data;

        if ((siWatCh >= ADC_CHANNEL_0) && (siWatCh < ADC_CHANNEL_MAX))
        {
            SAL_TaskSleep(100);
        }
    }

    return adc_scan_val;
}
```

Note: Refer to sources/dev.driver/adc/tcc70xx/adc.c

8 REFERENCES

[1] TCC70xx Full Specification

[2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

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